

UPMANYU JHA

AI/Machine Learning Engineer – Agentic AI · Production GenAI · RAG · MLOps

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SUMMARY

AI/ML Engineer with 3 years of engineering experience building production-grade agentic and LLM systems (RAG, fine-tuning, evaluation, LLM lifecycle management) on Google Cloud Platform (GCP) in regulated financial services, with end-to-end delivery ownership, client-facing solutioning and governance

EXPERIENCE

Quantiphi Analytics Solutions Private Limited (Google Cloud Partner)

Mumbai, IN

Machine Learning Engineer - Pod Lead (LLMs, RAG, Gen AI, Agentic AI)

Jan 2024 - Present

- Boosted document classification accuracy by 19.5% by redesigning the extraction and retrieval pipeline into reusable microservices on GCP and AWS, processing 500K+ docs/month across 5+ regulated business units and reducing manual review 80%
- Raised financial document extraction F1-score from 0.55 to 0.92 (+37 points) by building a RAG pipeline with vectorization and prompt-chained multi-agent orchestration over compliance-sensitive data, cutting audit time 40%
- Reduced LLM hallucinations by 85% through SFT and PEFT fine-tuning grounded with Document AI, serving inference via REST and gRPC service contracts and lowering cloud costs by 15%
- Built event-driven ingestion for 200K+ docs/month by architecting async Kafka-style streaming on Pub/Sub, ensuring auditable, reliable processing (~30 sec/doc) for production LLM workloads in a regulated environment
- Enabled safe, zero-downtime rollouts across 3 LLM services by implementing Kubernetes model lifecycle management with a model registry, pinned aliases, programmatic version detection, and rollback
- Increased fraud detection recall by 35% (precision >90%) by engineering a financial anomaly detection model using statistical profiling and predictive modeling, cutting review queues 50% with continuous drift monitoring
- Drove the technical win of an **\$85K sub-company engagement** – requested by the client as their architect-level advisor – partnering with GTM and client senior architects on architecture review, risk analysis, timeline, and resource-mix estimates
- Served as the client's long-term technical advisor on **\$150K multi-phase DocAI and Agentic AI workstreams** over 2 years, retained by client request for roadmap ideation after Phase 2 and Phase 3 use cases reached pilot-ready maturity
- Staffed and launched the new engagement by interviewing and onboarding two senior MLEs and a senior SDE, then leading knowledge transfer to kick off the sub-company's first four use cases
- **Leadership & Delivery:** Shipped 15+ E2E GenAI pipelines to production on a 12-month roadmap by leading a cross-functional squad of 8+ engineers across 5+ business units, enforcing testing and code-review governance
- Mentored 3 junior ML engineers through weekly code reviews and fine-tuning workshops, and conducted technical interviews for internal/external project hiring for senior GenAI and DocAI based roles

Machine Learning Engineer (Computer Vision, NLP)

Jul 2023 - Jan 2024

- Delivered peptide structure prediction with AlphaFold for a pharmaceutical Phase-1 trial, earning a company-wide Think-Tank innovation award
- Automated vehicle identification across warehouse CCTV feeds by fine-tuning YOLOv8 to detect vertically-oriented number plates under low-angle, variable-lighting conditions, achieving 86% detection accuracy

Bugcrowd

Mumbai, IN

Security Researcher | Bug Bounty Hunter

Jun 2021 - Jul 2023

- Improved web and API security testing throughput ~60% by developing automation workflows in Python and Go across large application surfaces in security-critical environments
- Reduced manual review overhead by integrating automated compliance checks into CI/CD pipelines, reinforcing audit-ready engineering practices
- Strengthened application hardening by performing RCA on 30+ architectural vulnerabilities and leading log analysis, improving incident response across event-driven service architectures

PROJECTS AND RESEARCH

AstraShield - Disaster Relief Agentic Verification System (MCP + HITL) | 1st Place, Q's Internal Round - Mumbai Hacks 2025

- Achieved ~90% detection accuracy by building an agentic Q&A and verification system with MCP-based function calling and human-in-the-loop review, using FastAPI, Gemini Vision, and React

Tools Used: Python, FastAPI, Gemini Vision, Model Context Protocol (MCP), React, GCP

Financial Filings RAG Pipeline - SEC EDGAR 10-K & ESG Intelligence

- Achieved sub-2s semantic retrieval over unstructured financial filings by deploying containerized async endpoints with pgvector storage and an LLM-as-judge evaluation pipeline, using Python, FastAPI, Vertex AI, and RAGAS

Tools Used: Python, FastAPI, AlloyDB (pgvector) / FAISS, Vertex AI / Hugging Face, RAGAS, Docker, GitHub Actions

Kisan-Setu - AWS Production Deployment (Textract + Bedrock + Lambda + SageMaker)

- Cut long-context prompting costs ~40% by architecting an offline-first serverless backend (DynamoDB, AppSync) and multimodal WhatsApp AI agent using AWS Bedrock and Textract to digitize rural agricultural operations for clusters of 500+ farmers

Tools Used: AWS Bedrock, Textract, SageMaker, Lambda, DynamoDB, CloudWatch, AppSync

EDUCATION

Bachelor of Engineering in Computer Engineering | University Of Mumbai, Mumbai, IN

Jul 2019 - May 2023

TECHNICAL SKILLS

Languages: Python, Go, C++, SQL
Agentic AI & GenAI: LLMs (Claude API, Claude Code, Gemini, OpenAI), RAG, Multi-Agent Systems, MCP, A2A, ADK, LangChain, Prompt Engineering, LLM Evaluation (RAGAS, LLM-as-judge), Hallucination Mitigation, HITL
ML: PyTorch, TensorFlow, Scikit-learn, Hugging Face, Vector DBs (FAISS, Pinecone)
Cloud & MLOps: AWS (Bedrock, SageMaker, Lambda, Textract), GCP (Vertex AI, Gemini Enterprise, Agent Platform, BigQuery), Kubernetes, Docker, CI/CD (GitHub Actions, pytest), Model Registry & Versioning
Distributed Systems & APIs: Event-Driven Architecture, Kafka, Pub/Sub, gRPC, Message Queues

CERTIFICATIONS AND TRAINING

Google Cloud Gemini Enterprise Development (GCP GED) Google Cloud	July 2026
Google Cloud Gemini Enterprise Agent Development (GCP GEAD) Google Cloud	July 2026
Google Generative AI Leader (GCP Gen Ai) Google Cloud	Dec 2025
Google Cloud Professional Cloud Architect (GCP PCA) Google Cloud	Jun 2025
Google Cloud Professional Machine Learning Engineer (GCP PMLE) Google Cloud	May 2024

AWARDS AND RECOGNITION

- **SPOT Award, Q2 2026** – sole quarterly winner: led an enterprise client's DocAI workstream as acting lead and primary client contact, owning delivery, risk mitigation, and team upskilling
- **Think-Tank Award (Innovative Thinking), Q1 2026** – sole global winner for redesigning the delivery approach on a DocAI engagement
- **Kaizen “Fish Eye” Award, Q1 2026** – recognized for first-time-right engineering quality with negligible production defects
- **Think-Tank Award, H1 2024** – AlphaFold peptide structure prediction supporting a pharmaceutical Phase-1 trial